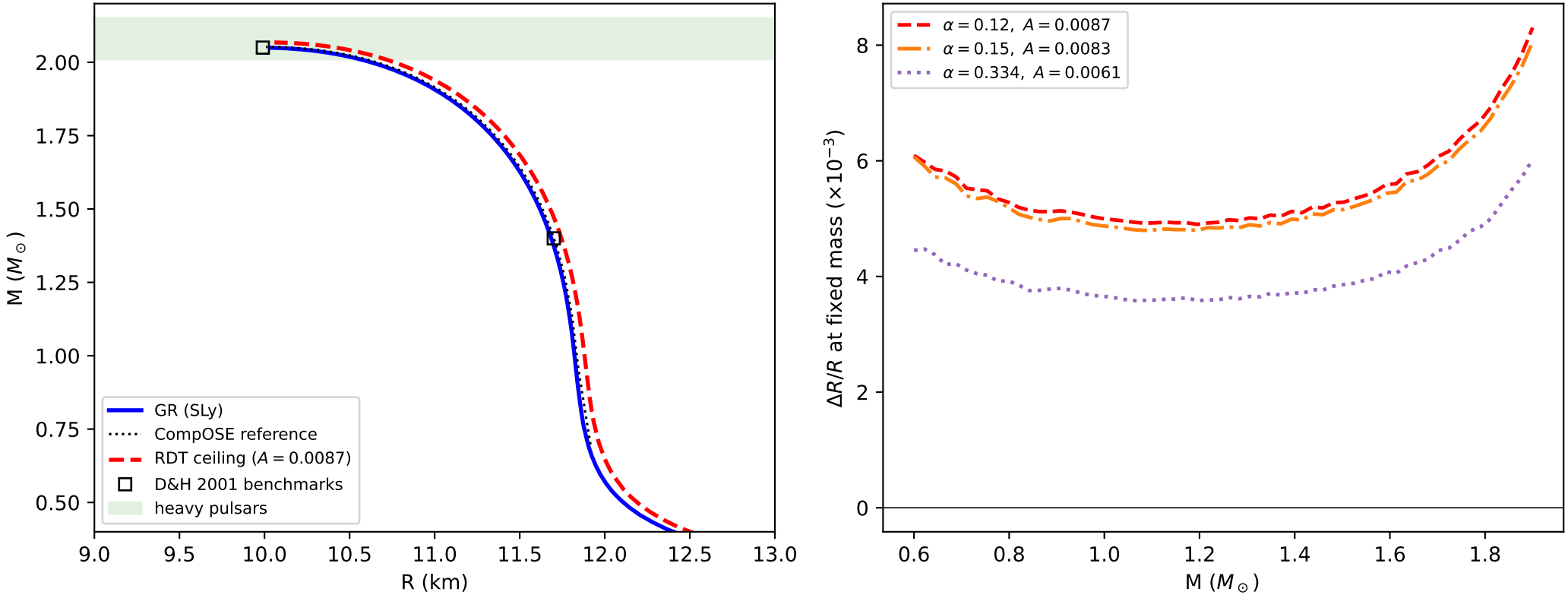
(a) Mass–radius: GR vs maximum solar-allowed RDT
M ($M_\odot$)
R (km)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
9.0
9.5
10.0
10.5
11.0
11.5
12.0
12.5
13.0
GR (SLy)
CompOSE reference
RDT ceiling ($A = 0.0087$)
D&H 2001 benchmarks
heavy pulsars
(b) Radius shift along the solar exclusion boundary
$\Delta R/R$ at fixed mass ($\times 10^{-3}$)
M ($M_\odot$)
8
6
4
2
0
0.6
0.8
1.0
1.2
1.4
1.6
1.8
$\alpha = 0.12,\ A = 0.0087$
$\alpha = 0.15,\ A = 0.0083$
$\alpha = 0.334,\ A = 0.0061$